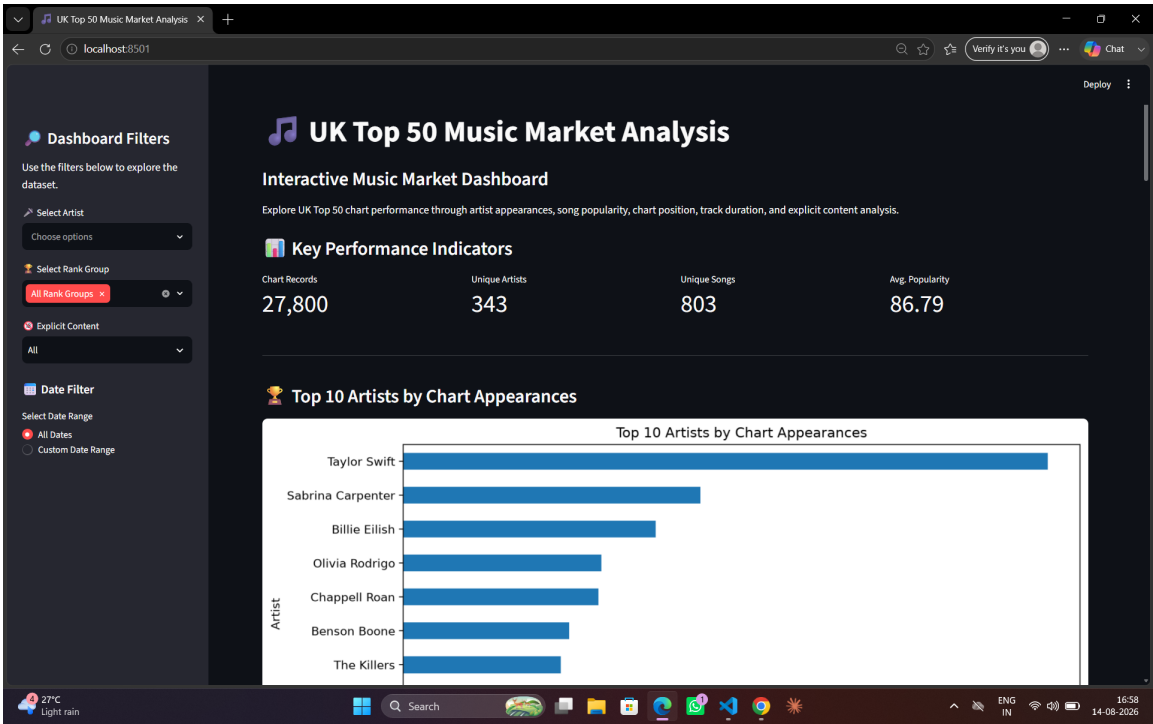


UK Top 50 Music Market Analysis

Data Analytics Project Report

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Tools: Python | Pandas | NumPy | Matplotlib | Streamlit | Jupyter Notebook



Interactive analysis of UK Top 50 chart performance, artist concentration, popularity, track duration, explicit content, and market diversity.

1. Project Overview

The UK Top 50 Music Market Analysis project examines music chart data to identify patterns in artist performance, song popularity, chart position, track duration, collaborations, album characteristics, and explicit content. The project combines exploratory data analysis with an interactive Streamlit dashboard so that the dataset can be explored dynamically.

Project Objectives

- Analyze artist dominance and repeated chart appearances.
- Examine the relationship between chart position and popularity.
- Compare average track duration across chart rank groups.
- Analyze explicit and non-explicit content distribution.
- Identify highly popular songs and overall market diversity.
- Build an interactive dashboard with artist, rank group, explicit-content, and date filters.

2. Dataset Overview

The final dashboard dataset contains 27,800 chart records representing 343 unique artists and 803 unique songs. The average popularity score across the analyzed records is 86.79.

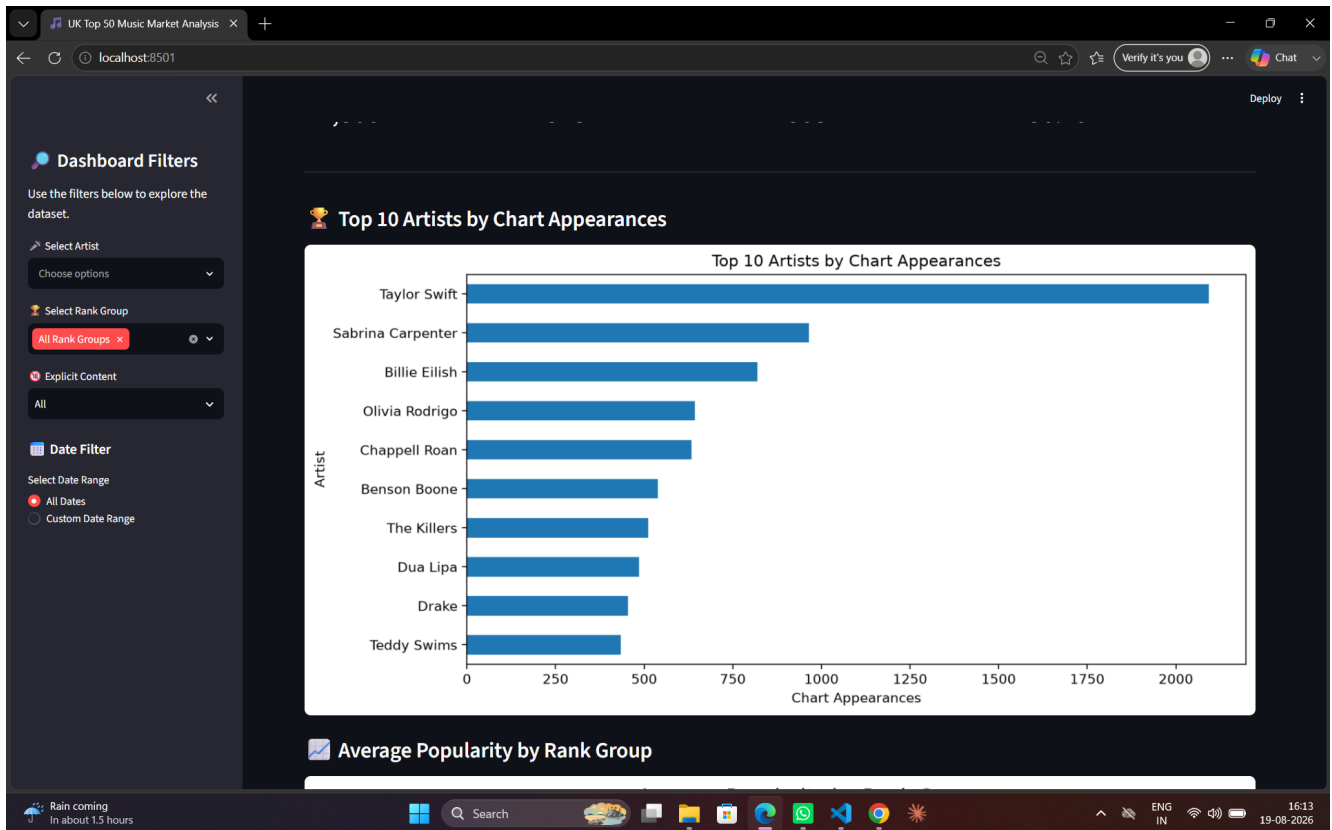
KPI	Result
Total Chart Records	27,800
Unique Artists	343
Unique Songs	803
Average Popularity	86.79

3. Tools and Technologies

Python was used for analysis and application logic; **Pandas** and **NumPy** for data manipulation; **Matplotlib** for visualization; **Streamlit** for the interactive dashboard; **Jupyter Notebook** for exploratory analysis; and **Git/GitHub** for version control and project publishing.

4. Top Artists by Chart Appearances

Artist appearances were counted to identify performers with a strong and repeated presence in the UK Top 50. Taylor Swift recorded the highest number of appearances in the analyzed dataset.



Top 10 artists by chart appearances.

5. Popularity by Rank Group

Chart positions were grouped into Top 10, positions 11-25, and positions 26-50. The Top 10 group has the highest average popularity (about 90.72), followed by 11-25 (about 87.08) and 26-50 (about 85.05). This indicates a relationship between chart position and popularity.



Average popularity across the three chart rank groups.

6. Track Duration Analysis

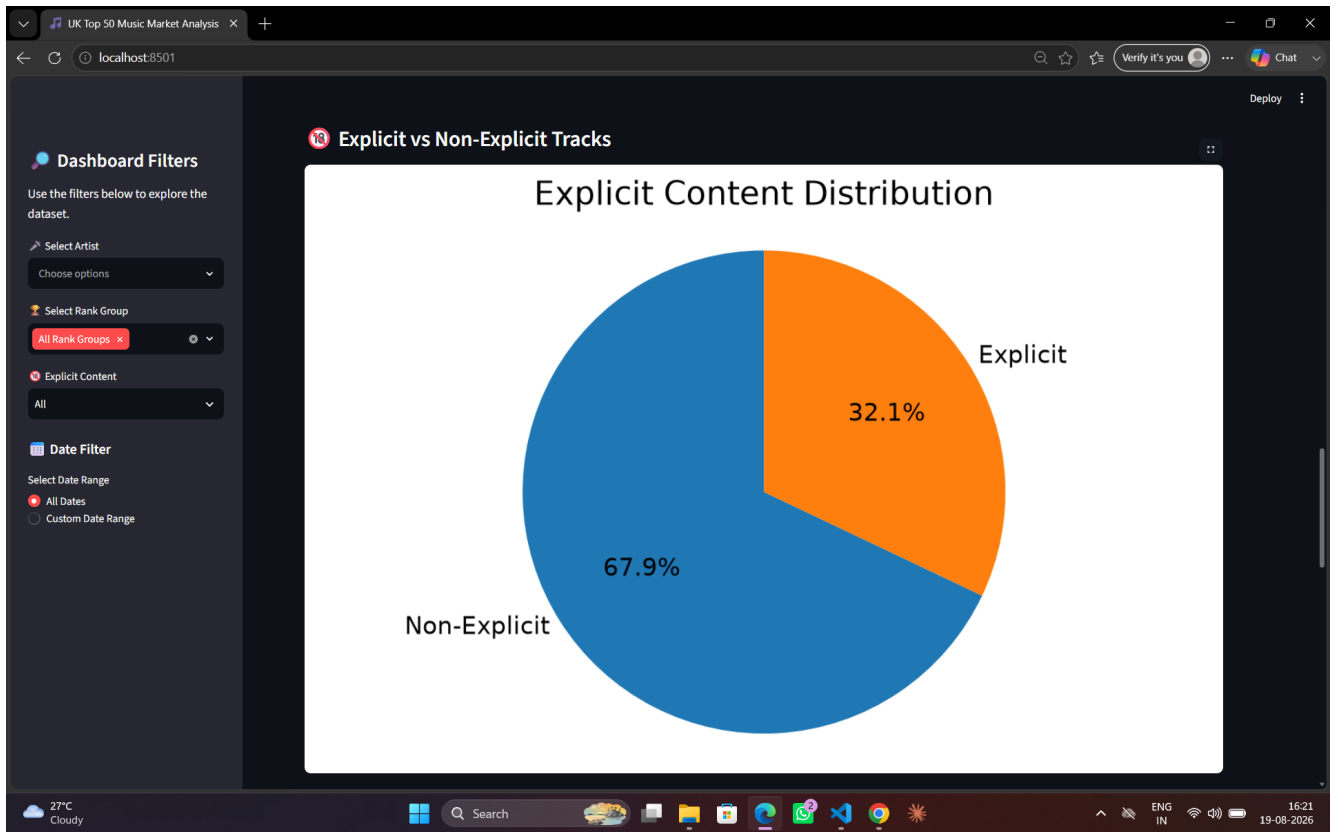
Average track duration remains relatively similar across rank groups: approximately 3.23 minutes for the Top 10, 3.24 minutes for positions 11-25, and 3.34 minutes for positions 26-50. Track length alone therefore does not appear to be a major differentiator of chart performance.



Average track duration by chart rank group.

7. Explicit Content Analysis

Non-explicit tracks represent the majority of analyzed chart records. Approximately 67.9% of records are non-explicit and 32.1% are explicit.



Explicit versus non-explicit content distribution.

8. Top Songs by Popularity

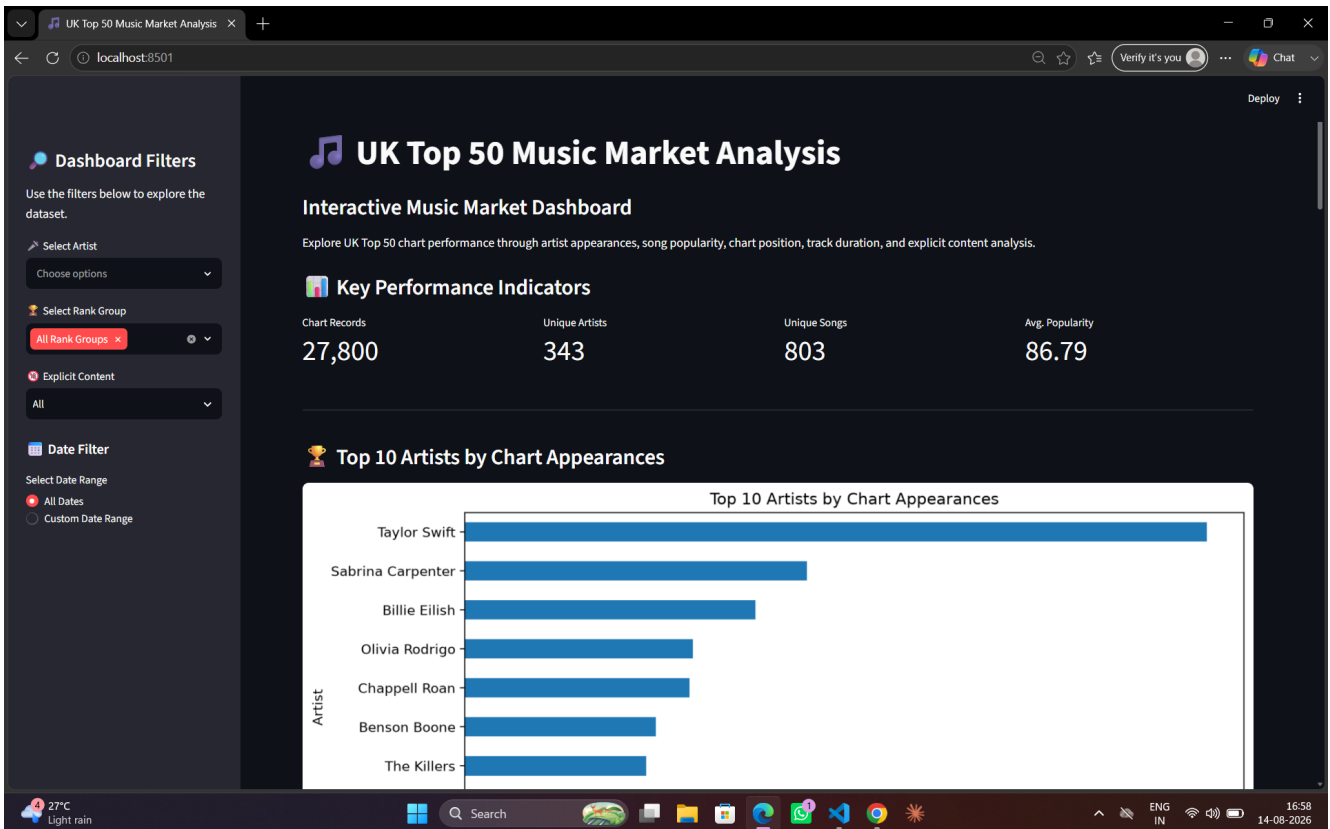
The dashboard ranks highly popular chart records using the popularity score and displays song, artist, popularity, and chart position. This provides a quick view of highly popular music represented in the dataset.

🎵 Top 10 Songs by Popularity			
song	artist	popularity	position
Die With A Smile	Lady Gaga & Bruno Mars	100	13
Die With A Smile	Lady Gaga & Bruno Mars	100	11
Die With A Smile	Lady Gaga & Bruno Mars	100	15
i like the way you kiss me	Artemas	100	3
Cruel Summer	Taylor Swift	100	6
i like the way you kiss me	Artemas	100	3
BIRDS OF A FEATHER	Billie Eilish	100	2
All I Want for Christmas Is You	Mariah Carey	100	2
greedy	Tate McRae	100	13
Die With A Smile	Lady Gaga & Bruno Mars	100	7

Top songs by popularity as displayed in the dashboard.

9. Interactive Streamlit Dashboard

The Streamlit dashboard provides an interactive interface for exploring the project. Users can filter the dataset by artist, rank group, explicit-content status, and date range. The KPIs, charts, and tables update according to the selected filters.



Dashboard overview with filters and KPI cards.

10. Business Insights

- 1. Strong Artist Concentration:** A relatively small number of artists account for a large number of repeated chart appearances, with Taylor Swift leading the analyzed dataset.
- 2. Popularity and Chart Position:** Top 10 tracks have the highest average popularity compared with lower rank groups.
- 3. Track Duration:** Average duration is similar across rank groups, so song length alone is not a major differentiator of chart performance.
- 4. Explicit Content:** Non-explicit tracks form the majority of analyzed records.
- 5. Market Diversity:** Hundreds of artists and songs are represented, demonstrating broad diversity in the analyzed UK Top 50 market.

Business Insights

1. Strong Artist Concentration

A relatively small number of artists account for a large number of repeated chart appearances. Taylor Swift has the highest number of appearances in the analyzed dataset.

2. Popularity and Chart Position

Top 10 tracks have the highest average popularity compared with lower rank groups, indicating a relationship between chart position and popularity.

3. Track Duration

Average track duration remains relatively similar across the different rank groups, suggesting that track length alone is not a major differentiator of chart performance.

4. Explicit Content

Non-explicit tracks represent the majority of records in the analyzed dataset.

5. Market Diversity

The dataset contains hundreds of artists and songs, demonstrating a broad range of music represented in the UK Top 50 market.

Business insights presented in the dashboard.

11. Conclusion

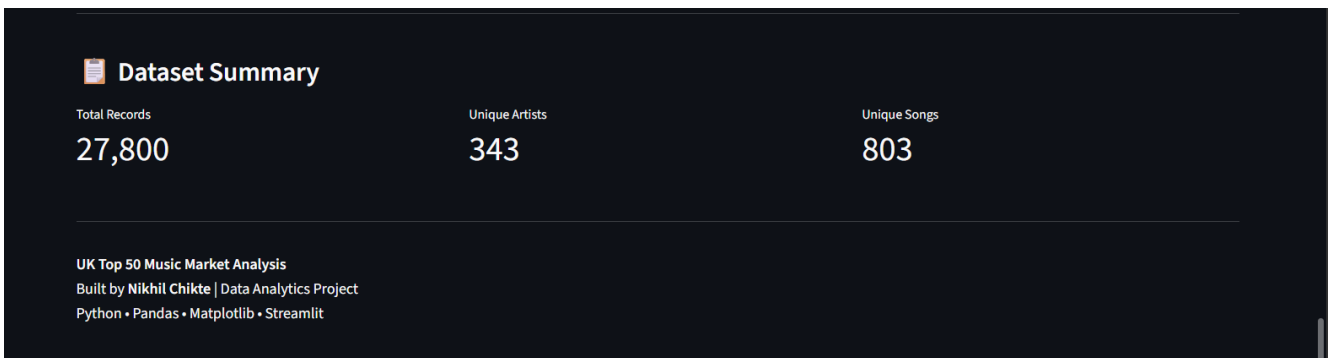
The UK Top 50 Music Market Analysis demonstrates a complete data analytics workflow: data preparation, exploratory analysis, visualization, business insight generation, interactive dashboard development, version control, and cloud deployment. The results show strong artist concentration, higher popularity among Top 10 tracks, relatively stable track duration across rank groups, a majority of non-explicit records, and broad artist and song diversity.

12. Project Links

GitHub Repository: <https://github.com/nikhilchikte376-hash/UK-Top-50-Music-Market-Analysis>

Live Streamlit Dashboard:

<https://uk-top-50-music-market-analysis-ohdnqobw8ume7lhywkchom.streamlit.app/>



Final dataset summary and project footer.